



Automated heuristic discovery for solving combinatorial optimization problems

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1 Introduction

<TODO: description of the topic of the doctoral thesis - including a clear research question>

Deep neural networks (DNNs) can be used to solve a variety of optimization problems, with a particular focus on vehicle routing problems (VRPs) in recent literature. There has been great progress with regard to problem sizes considered, VRP variants studied, and ever-improving architectures and methods proposed [1], [2], [3], [4]. However, the literature has primarily focused on problems for which it is trivial to find feasible solutions. This stands in stark contrast to practical applications, where finding feasible solutions may be difficult due to restrictive real-world constraints.

In a real vehicle routing problem with time windows, stochastic times, and heterogeneous vehicle fleets (w.r.t. their capacity or skills/services they can offer), just finding any feasible solution can be a major challenge. ML techniques need to overcome this if they are ever to see widespread use. Therefore, this research project will examine challenging combinatorial optimization (CO) problems where feasible solutions are non-trivial to generate and develop novel, DRL- and LLM-based techniques for solving them.

<TODO: include something on LLM research>

<TODO: This introduction is supposed for the entire PhD, so it should have an overarching contextualization of the different topics/projects of my PhD.>

2 Related work

<TODO: outline of the state of research, relevant literature>

3 Research projects

<TODO: illustration of the chosen research methods>

3.1 RouteFinder: Towards Foundation Models for Vehicle Routing Problems

<TODO: summary of the paper, and how it relates to the research project>

3.2 Learning to Solve Highly Constrained Routing Problems with Deep Reinforcement Learning

<TODO: summary of the paper, and how it relates to the research project>

<TODO: mention the instance generator!>

3.3 LLM-Driven Heuristic Discovery for Highly-Constrained Routing Problems

<TODO: outline of the third research project>

4 Choice of supervisor(s)

<TODO: argue for Kevin>

5 Remaining Project Plan

<TODO: time and work schedule, and possibly financial budget and overview of resources>

References

- [1] I. Bello, H. Pham, Q. V. Le, M. Norouzi, and S. Bengio, “Neural Combinatorial Optimization with Reinforcement Learning,” in *5th International Conference on Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Workshop Track Proceedings*, OpenReview.net, 2017.
- [2] W. Kool, H. van Hoof, and M. Welling, “Attention, Learn to Solve Routing Problems!,” in *International Conference on Learning Representations*, 2019.
- [3] Y.-D. Kwon, J. Choo, B. Kim, I. Yoon, Y. Gwon, and S. Min, “POMO: Policy Optimization with Multiple Optima for Reinforcement Learning,” in *Advances in Neural Information Processing Systems*, Curran Associates, Inc., 2020, pp. 21188–21198.
- [4] A. Hottung, P. Wong-Chung, and K. Tierney, “Neural Deconstruction Search for Vehicle Routing Problems,” *Transactions on Machine Learning Research*, 2025.